Ch 4.3.3 and 4.3.4 - Multiple and Multinomial Logistic Regression

Lecture 11 - CMSE 381

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Announcements

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1	
6	Wed	9/11	Multi-linear regression	3.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due Dun 9/15
8	Mon	9/16	Last of the linear regression		
9	Wed	9/18	Intro to classification, Bayes classifier, KNN classifier	2.2.3	
10	Fri	9/20	Logistic Regression	4.1, 4.2, 4.3.1-3	Hw #3 Due Sun 9/22
11	Mon	9/23	Multiple Logistic Regression / Multinomial Logistic Regression	4.3.4-5	
	Wed	9/25	***Project Day & Review***		
	Fri	9/27	**Midterm #1**		

Announcements:

- Wednesday - Project day
  - Send me a message or email if you're planning on doing an honors version of the project.
- Wednesday - Review day
  - Nothing prepped
  - Bring your questions
- Friday - Exam #1
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in
  - Non-internet calculator if you want it

Covered in this lecture

Last Time:

- Logistic Regression

This time:

- Multiple Logistic Regression
- Multinomial Logistic Regression

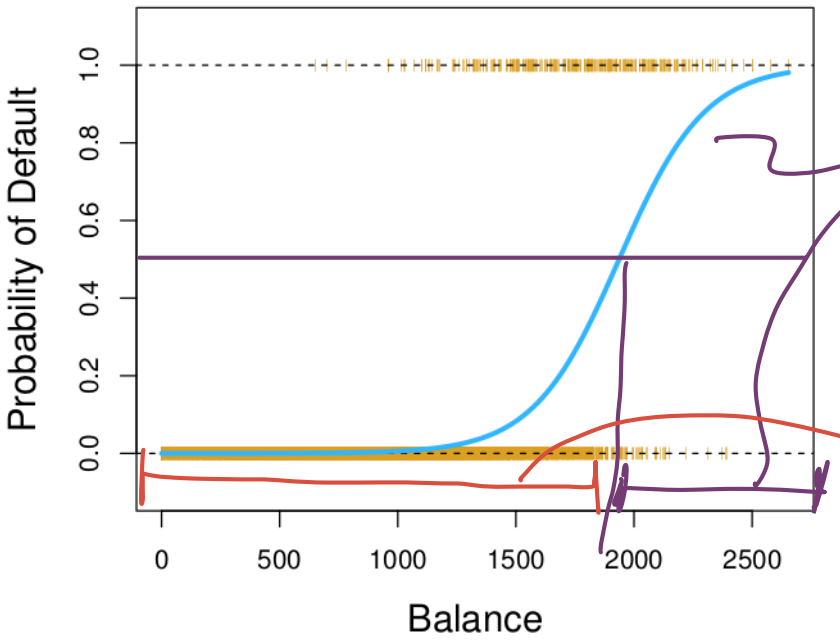
Section 1

Review of Logistic Regression from last time

Logistic regression

- Assume single input $\mathtt{X}$
- Output takes values $\mathtt{Y} \in \{\mathtt{Yes}, \mathtt{No}\}$

$$p(\mathtt{X}) = \Pr(\mathtt{Y} = \mathtt{yes} \mid \overset{X}{\cancel{\mathtt{balance}}})$$



higher prob of yes than no
so predict yes

$$p(\mathtt{x}) = \frac{e^{\beta_0 + \beta_1 \mathtt{x}}}{1 + e^{\beta_0 + \beta_1 \mathtt{x}}}$$

lower prob of
yes than no
$\Rightarrow$ predict no

How to get logistic function

Assume the (natural) log odds (logits) follow a linear model

$$\log\left(\underbrace{\boxed{\frac{p(x)}{1-p(x)}}}_{\text{odds}}\right) = \beta_0 + \beta_1 x$$

Solve for $p(x)$:

$$p(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

Playing with the logistic function: desmos.com/calculator/cw1pyzzqci

Section 2

Multiple Logistic Regression

New assumption

$p \geq 1$ input variables

$X_1, X_2, \cdots, X_p$

Y output variable has only two levels

$Y \in \{Yes, No\}$

$\in \{0, 1\}$

Multiple Logistic Regression

Multiple features:

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

Equivalent to:

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Example

	default	student (X_1)	balance (X_2)	income (X_3)
0	No	No	729.526495	44361.625070
1	No	Yes	817.180407	12106.134700
2	No	No	1073.549164	31767.138950
3	No	No	529.250605	35704.493940
4	No	No	785.655883	38463.495880
5	No	Yes	919.588531	7491.558572
6	No	No	825.513331	24905.226580
7	No	Yes	808.667504	17600.451340
8	No	No	1161.057854	37468.529290
9	No	No	0.000000	29275.268290

⋮

Default data set

Yes

Predict `default` from `balance`, `student`, and `income`

$$p(x) = \Pr(Y = \text{yes} \mid X = (x_1, x_2, x_3)) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3}}$$

Section 3

Multinomial Logistic Regression

New assumption

$p \geq 1$ input variables

$X_1, X_2, \cdots, X_p$

Y output variable has K levels

$Y \in \{\text{blue}, \text{green}, \text{pink}, \text{orange}\}$

$K = 4$

Remember dummy variables?

Slide from linear regression days

`Region`:

	x_{i1}	x_{i2}
South	1	0
West	0	1
East	0	0

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

Baseline = the level not used

= East

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

Example

Predict $Y \in \{\texttt{stroke}, \texttt{overdose}, \texttt{seizure}\}$ for hospital visits based on some input(s) X

example

$$\Pr(Y = \texttt{stroke} \mid X = x) = 1/4$$

$$\Pr(Y = \texttt{overdose} \mid X = x) = 1/2$$

$$\Pr(Y = \texttt{seizure} \mid X = x) = 1 - \left(\tfrac{1}{4} + \tfrac{1}{2}\right) = 1/4$$

can figure out third from the other two.

Example: predict highest prob outcome

here, overdose for this particular x

Multinomial Logistic Regression

Plan A

little k 1, 2, —, K−1

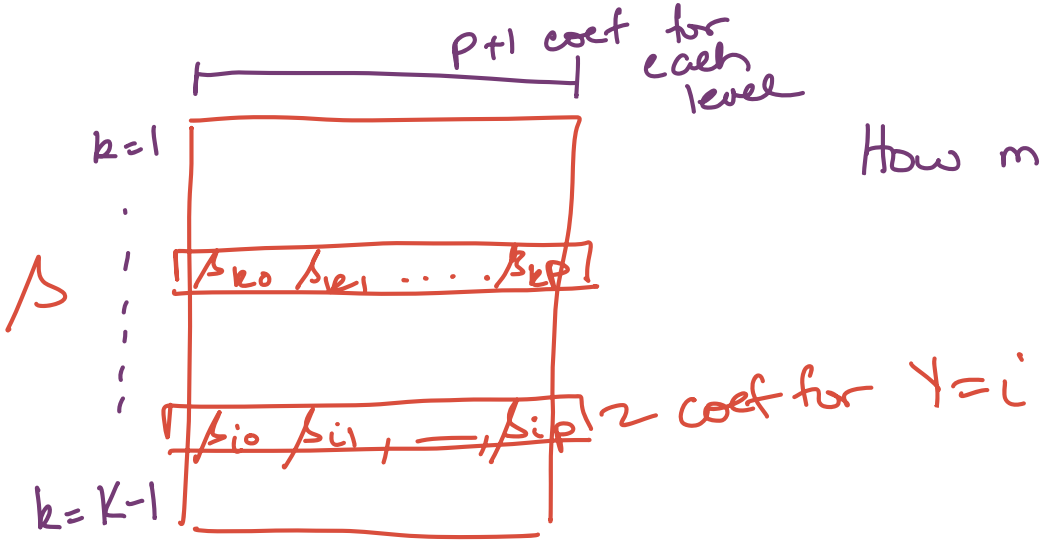
$\beta_{k0}, \ldots, \beta_{kp}$ coef for that k

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

Same for every k

$$\Pr(Y = K|X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

- Assume Y has K levels
- Make K (the last one) the baseline



How many β coefficients?

$(K-1)(p+1)$

Example

Predict $Y \in \{\texttt{stroke}, \texttt{overdose}, \texttt{seizure}\}$ for hospital visits based on Xp

$$\Pr(Y = \texttt{stroke} \mid X = x) = \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x) = \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x) = \frac{1}{1 + \exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}$$

$\beta_{str,0}$ $\beta_{str,1}$

$\beta_{od,0}$ $\beta_{od,1}$

~~$\beta_{sez,0}$ $\beta_{sez,1}$~~

Log odds

Calculated so that log odds between *any pair of* classes is linear.
Specifically, for $Y = k$ vs $Y = K$, we have

$k \in \{1, 2, \ldots, K-1\}$ Baseline

$$\log\left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)}\right) = \beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p$$

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

$$\Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

Plan B: Softmax coding

Treat all levels symmetrically

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{\sum_{l=1}^{K} e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}.$$

$$\beta = \begin{bmatrix} \beta_{10} & \beta_{11} & \cdots & \beta_{1p} \\ \vdots & & & \vdots \\ \beta_{k0} & \beta_{k1} & \cdots & \beta_{kp} \\ \vdots & & & \vdots \\ \beta_{K0} & \beta_{K1} & \cdots & \beta_{Kp} \end{bmatrix}$$

$K \times (p+1)$ coeff.

Calculated so that log odds between two classes is linear

$$\log\left(\frac{\Pr(Y = k|X = x)}{\Pr(Y = k'|X = x)}\right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1} - \beta_{k'1})x_1 + \cdots + (\beta_{kp} - \beta_{k'p})x_p.$$

Used version used often in ML contexts

Resulting learned model makes the same predictions.

Softmax example

$$\Pr(Y = \texttt{stroke} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{overdose} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

$$\Pr(Y = \texttt{seizure} \mid X = x)$$
$$= \frac{\exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}{\exp(\beta_{\texttt{str},0} + \beta_{\texttt{str},1}x) + \exp(\beta_{\texttt{OD},0} + \beta_{\texttt{OD},1}x) + \exp(\beta_{\texttt{seiz},0} + \beta_{\texttt{seiz},1}x)}$$

Jupyter Notebook

Next time

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